Kathmandu University · Department of Mathematics

Modern Algebra (MATH 402) · Course Project

Lattice-Based Encryption

A Post-Quantum Cryptosystem and its Algebraic Foundations

Sahaj Raj Malla [redacted] Roll No. [redacted]
Rumi Suwal [redacted] Roll No. [redacted]
Priti Tamang [redacted] Roll No. [redacted]

Instructor: Dr. Gokul KC

The Post-Quantum Problem

- Today's public-key encryption (RSA, elliptic curves) rests on factoring and discrete logarithms.
- Shor's algorithm solves both efficiently on a large quantum computer. That breaks RSA and ECC.
- Harvest now, decrypt later: encrypted traffic can be stored today and opened once such a machine exists.
- We need hardness that quantum computers do not undo. Lattices are one of the leading answer.

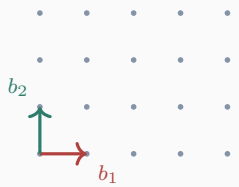
What is a Lattice?

Definition

For a basis $b_1, \ldots, b_n \in \mathbb{R}^n$,

$$\mathcal{L}(B) = \left\{ \sum_{i=1}^{n} a_i b_i : a_i \in \mathbb{Z} \right\}.$$

- Simplest case: $(1, 0)$ and $(0, 1)$ generate $\mathbb{Z}^2$.
- Integer steps along the basis give a regular grid of points.
- Real cryptographic lattices live in 1024 dimensions, but the core picture never changes: a basis, integer steps, and a grid of dots.



A lattice in the plane

One Lattice, Many Bases



Good basis: short, orthogonal, private



Bad basis: long, skewed, public

- Both bases generate the *same* lattice.
- The basis you hold is the secret.
  - A good basis makes lattice problems easy
  - A bad one makes them hard
  - That gap is the key to lattice cryptography

The Hard Problems

Shortest Vector Problem (SVP)

Given a basis B, find a nonzero $v \in \mathcal{L}(B)$ of smallest norm.

Closest Vector Problem (CVP)

Given a basis B and a target t, find the lattice point nearest t.

- Both are NP-hard in the worst case.
- No efficient quantum algorithm is known. It survives Shor.

The Algebraic Bridge

Each piece of the cryptosystem is an algebraic object.

Algebraic object	**Role in the scheme**
Group	the lattice: a discrete subgroup of $\mathbb{R}^n$
Ring	arithmetic mod q: $\mathbb{Z}_q$ and $\mathbb{Z}_q[x]$
Quotient ring / ideal	$R_q = \mathbb{Z}_q[x]/(x^n + 1)$
Module	vectors over R_q (Module-LWE)

The Quotient Ring R_q

$$R_q = \mathbb{Z}_q[x] \,/\, (x^n + 1)$$

- Elements are polynomials of degree below n with coefficients in $\mathbb{Z}_q$.
- The relation $x^n + 1 \equiv 0 \implies \mathbf{x^n \equiv -1}$ folds higher powers back down.

How it folds (Just like $\mathbb{Z}_n$ wrapping at n)

If $n = 4$, the relation $x^4 \equiv -1$ forces all products to stay inside the ring:

$$x^4 \equiv \mathbf{-1} \quad \longrightarrow \quad x^5 \equiv \mathbf{-x} \quad \longrightarrow \quad x^6 \equiv \mathbf{-x^2}$$

- **The Payoff:** This compact arithmetic (e.g. $n = 256$ fits in 512 bytes) is exactly what makes Ring-LWE efficient.

Learning With Errors (LWE)

Search-LWE

Given many pairs (a, b) with $b = a \cdot s + e \pmod{q}$ and small random noise e, recover the secret s.

- Without the noise e, this is ordinary linear algebra: easy.
- With small noise, it is believed hard.

The Role of the Noise e

How e is generated

Each coefficient of e is drawn independently from a narrow distribution centered at 0, giving small integers such as $-1, 0, 1$. Real schemes use a discrete Gaussian or a centered binomial; our toy example uses ternary noise.

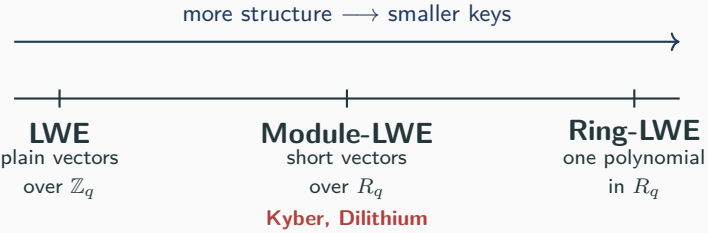
Why the noise makes it hard

- No noise: $b = a \cdot s$ is one invertible equation, so divide by a and read off s. Plain linear algebra.
- With noise: $b = a \cdot s + e$ sits just off the lattice point $a \cdot s$, so recovering s becomes the hard closest vector problem.
- Regev's Theorem: breaking average-case LWE is as hard as worst-case lattice problems. The noise is the security.

e must be large enough to hide s, yet small enough to round away in decryption.

From LWE to Ring-LWE

Adding algebraic structure shrinks the keys.



Module-LWE is the middle setting the standards chose: a tunable balance of structure and safety.

Ring-LWE Toy Example: The Ring

We work in

$$R_q = \mathbb{Z}_{17}[x] \,/\, (x^4 + 1), \qquad q = 17, \quad n = 4.$$

Folding rule

$x^4 \equiv -1$, and therefore

$$x^5 \equiv -x, \quad x^6 \equiv -x^2, \quad x^7 \equiv -x^3.$$

Every product of two degree-3 polynomials folds back to degree below 4.

Conventions

- Coefficients live in $\mathbb{Z}_{17}$.
- We write them centered in $\{-8, \ldots, 8\}$ so that "small" stays visible.
- $\lfloor q/2 \rfloor = 8$ will encode a message bit.

Roles: a public and random, s secret, e error.

Step 1: Key Generation

Pick a random a, a small secret s, and small noise e:

$$a = 4 + x + 11x^2 + 10x^3, \qquad s = 1 + x - x^3, \qquad e = -x + x^2.$$

Multiply $a \cdot s$, then fold with $x^4 \equiv -1,\ x^5 \equiv -x,\ x^6 \equiv -x^2$

$$\begin{aligned} a \cdot s &= 4 + 5x + 12x^2 + 17x^3 + 9x^4 - 11x^5 - 10x^6 \\ &= (4 - 9) + (5 + 11)x + (12 + 10)x^2 + 17x^3 \\ &= -5 + 16x + 22x^2 + 17x^3 \ \equiv\ 12 + 16x + 5x^2 \pmod{17}. \end{aligned}$$

Centered, $a \cdot s = -5 - x + 5x^2$, so the public key follows:

$$b = a \cdot s + e = -5 - 2x + 6x^2.$$

Public key (a, b), secret key s. The noise e is what stops anyone reading s straight out of b.

Step 2: Encryption

Encode the message bits $(1, 0, 1, 0)$ as $m = 1 + x^2$, so $\lfloor q/2 \rfloor\, m = 8 + 8x^2$. Draw small r, e_1, e_2:

$$r = -1 + x + x^3, \qquad e_1 = 1 - x^3, \qquad e_2 = x - x^2.$$

Two ciphertext polynomials (products folded as in Step 1)

$$u = a \cdot r + e_1 = 3 - 8x - 3x^2 + 4x^3,$$
$$v = b \cdot r + e_2 + \lfloor q/2 \rfloor\, m = -2 - 8x - x^2 + x^3.$$

The ciphertext is the pair (u, v). Each polynomial carries fresh noise, so the same message never encrypts the same way twice.

Step 3: Decryption

Compute $v - u \cdot s$. The shared term $a \cdot s \cdot r$ cancels, leaving

$$v - u \cdot s = \lfloor q/2 \rfloor \, m + \underbrace{(e \cdot r + e_2 - e_1 \cdot s)}_{\text{noise}}.$$

The numbers

$$v - u \cdot s = 7 + 6x^2 + 3x^3, \qquad \text{noise} = -1 - 2x^2 + 3x^3.$$

Every noise coefficient has $|\cdot| \leq 3 < q/4$, so it cannot flip a bit.

Round each coefficient to the nearer of 0 and 8

$$(7,\, 0,\, 6,\, 3) \longrightarrow (8,\, 0,\, 8,\, 0) \longrightarrow \text{bits } (1, 0, 1, 0).$$

Recovered message = original message.

CRYSTALS-Kyber, standardized as ML-KEM (FIPS 203)

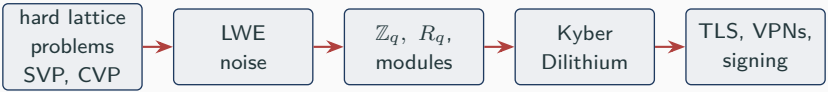
- A key encapsulation mechanism: two parties agree on a shared secret key. Replaces RSA and Diffie-Hellman key exchange.
- Public key is a noisy product $b = a \cdot s + e$ over modules in R_q.
- Encapsulate: add fresh noise to mask a random value. Decapsulate: the secret cancels the structure and the small noise rounds away.
- Parameters $n = 256$, $q = 3329$. The cost is size: a few thousand bytes, where elliptic curves use a few dozen.

CRYSTALS-Dilithium, standardized as ML-DSA (FIPS 204)

- Digital signatures: prove a message came from the key holder and was not altered. Replaces RSA and ECDSA.
- Security rests on Module-LWE plus Module-SIS: find a short nonzero combination of ring elements summing to zero.
- Signing and verifying use short vectors and a hash challenge; forging one would solve Module-SIS.
- About 125 bits of post-quantum security. FALCON, on NTRU lattices, is the smaller-signature alternative.

Deployment and the Whole Picture

- 2024: NIST standardizes ML-KEM and ML-DSA. Rollout into TLS, VPNs, and code signing is underway.
- Hybrid mode runs a classical and a lattice scheme together through the transition.



References

- O. Regev. *The Learning with Errors Problem (survey)*. 2010.
- V. Lyubashevsky, C. Peikert, O. Regev. *On Ideal Lattices and Learning with Errors over Rings*. 2010.
- C. Peikert. *A Decade of Lattice Cryptography*. 2016.
- NIST. *FIPS 203, 204, 205: Post-Quantum Cryptography Standards*. 2024.
- J. Hoffstein, J. Pipher, J. H. Silverman. *NTRU: A Ring-Based Public Key Cryptosystem*. 1998.

Thank you.

Questions?